

Evaluation of real Integrals by contour integration

A large number of real definite integrals, whose evaluation by usual method become sometime very tedious, can be easily evaluated by Cauchy's residue theorem.

Integral of type $\int_0^{2\pi} F(\cos\theta, \sin\theta) d\theta$

where $F(\cos\theta, \sin\theta)$ is a rational function of $\sin\theta$ and $\cos\theta$.

Such integrals can be converted to complex integrals by substituting $z = e^{i\theta}$ i.e. by considering a unit circle $|z|=1$, with center 0.

i.e. $z = r e^{i\theta} = 1 \cdot e^{i\theta}$

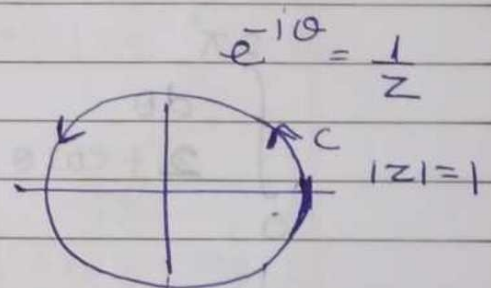
$$\boxed{z = e^{i\theta}}$$

$$z = e^{i\theta}$$

$$dz = i e^{i\theta} d\theta$$

$$\ast \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

$$\ast \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} = \frac{1}{2i} \left(z - \frac{1}{z} \right)$$



as θ , varies from 0 to 2π , z moves around unit circle in anticlockwise direction

$$\int_0^{2\pi} F(\cos \theta, \sin \theta) d\theta = \oint_{C: |z|=1} F\left(\frac{1}{2}\left(z + \frac{1}{z}\right), \frac{1}{2i}\left(z - \frac{1}{z}\right)\right) \frac{dz}{iz}$$

thus the integral on the right can be computed using residue theorem.

Q-1 →

Evaluate $\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta}$

Let $z = e^{i\theta}$ $\cos \theta = \frac{1}{2}\left(z + \frac{1}{z}\right)$
 $dz = i e^{i\theta} d\theta$
 $\Rightarrow d\theta = \frac{dz}{i e^{i\theta}} = \frac{dz}{iz}$

$$\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta} = \oint_{C: |z|=1} \frac{1}{2 + \frac{1}{2}\left(z + \frac{1}{z}\right)} \frac{dz}{iz}$$

$$= \oint_C \frac{1}{(z^2 + 4z + 1)} \cdot \frac{2}{i} dz$$

$$= \frac{2}{i} \oint_C \frac{1}{(z^2 + 4z + 1)} dz, \quad C: |z|=1$$

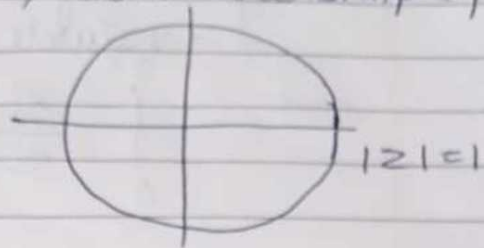
$$I = \frac{2}{i} I_1 \quad \text{--- (1)}$$

poles are

$$z = \frac{-4 \pm \sqrt{16 - 4}}{2} = -2 \pm \sqrt{3}$$

poles are $\alpha = -2 + \sqrt{3}$, $\beta = -2 - \sqrt{3}$, both are simple poles

only $z = -2 + \sqrt{3}$, lies inside
 $|z| = 1$



Residue at $z = -2 + \sqrt{3} = \alpha$

$$\text{Res. } f(z) = \lim_{z \rightarrow \alpha} (z - \alpha) \cdot f(z)$$

$$= \lim_{z \rightarrow \alpha} (z - \alpha) \frac{1}{z^2 + 4z + 1} \quad \left(\frac{0}{0} \right)$$

$$= \lim_{z \rightarrow \alpha} \frac{1}{2z + 4} = \frac{1}{2\alpha + 4}$$

$$= \frac{1}{2(-2 + \sqrt{3}) + 4} = \frac{1}{2\sqrt{3}}$$

by residue theo

$$I_1 = \oint_C \frac{1}{z^2 + 4z + 1} dz = 2\pi i \times \frac{1}{2\sqrt{3}} = \frac{\pi i}{\sqrt{3}}$$

from (i) $I = \frac{2}{i} I_1$

$$\begin{aligned} \int_0^{2\pi} \frac{d\theta}{2 + \cos\theta} &= \frac{2}{i} \oint_C \frac{1}{z^2 + 4z + 1} dz \\ &= \frac{2}{i} \times \frac{\pi i}{\sqrt{3}} = \frac{2\pi}{\sqrt{3}} \end{aligned}$$

Q.E.D.

Q1- } using complex variable techniques
evaluate the real integral

$$\int_0^{2\pi} \frac{\sin^2 \theta}{5-4\cos \theta} d\theta$$

Soln

$$z = e^{i\theta}$$

$$|z|=1$$

$$\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

$$\sin \theta = \frac{1}{2i} \left(z - \frac{1}{z} \right)$$

$$I = \int_0^{2\pi} \frac{\sin^2 \theta}{5-4\cos \theta} d\theta$$

$$dz = i e^{i\theta} d\theta$$

$$d\theta = \frac{dz}{i z}$$

$$I = \frac{1}{2} \int_0^{2\pi} \frac{2 \sin^2 \theta}{5-4\cos \theta} d\theta = \frac{1}{2} \int_0^{2\pi} \frac{(1-\cos 2\theta)}{5-4\cos \theta} d\theta$$

$$I = \frac{1}{2} \text{ Real part of } \int_0^{2\pi} \frac{1 - e^{2i\theta}}{(10 - 8\cos \theta)} d\theta \quad \text{--- (1)}$$

$$\int_0^{2\pi} \frac{1 - e^{2i\theta}}{(10 - 8\cos \theta)} d\theta = \oint \frac{1 - z^2}{10 - 8 \left(z + \frac{1}{z} \right)} \frac{dz}{i z}$$

$$= \oint \frac{1 - z^2}{10 - 4 \left(z + \frac{1}{z} \right)} \frac{dz}{i z}$$

$$= \frac{1}{i} \oint \frac{1 - z^2}{(10z - 4z^2 - 4)} dz$$

$$= \frac{1}{2i} \oint \frac{1 - z^2}{(5z - 2z^2 - 2)} dz$$

poles are $(2z-1)(z-2)=0 \Rightarrow z=2, \frac{1}{2}$

only $z = \frac{1}{2}$ lies inside $|z|=1$

$$\text{i.e. } \int_0^{2\pi} \frac{1 - e^{2i\theta}}{(10 - 8 \cos \theta)} d\theta = \frac{1}{2i} \oint_C \frac{1 - z^2}{(2z-1)(z-2)} dz$$

$$= \frac{1}{2i} (2\pi i \times \text{Residue at } z = \frac{1}{2})$$

$$= \frac{1}{2i} (\text{Residue at } z = \frac{1}{2})$$

$$\text{Res at } (z = \frac{1}{2}) = \lim_{z \rightarrow \frac{1}{2}} (z - \frac{1}{2}) \frac{1 - z^2}{(2z-1)(z-2)}$$

$$= \frac{1}{4}$$

$$\int_0^{2\pi} \frac{1 - e^{2i\theta}}{10 - 8 \cos \theta} d\theta = \frac{1}{2i} (2\pi i \times \frac{1}{4})$$

$$= \pi/4$$

$$\int_0^{2\pi} \frac{\sin^2 \theta}{5 - 4 \cos \theta} d\theta = \text{Real part of } \int_0^{2\pi} \frac{1 - e^{2i\theta}}{10 - 8 \cos \theta} d\theta$$

$$\boxed{\int_0^{2\pi} \frac{\sin^2 \theta}{5 - 4 \cos \theta} d\theta = \frac{\pi}{4}}$$

(3) Using contour integration evaluate

$$\int_0^{\pi} \frac{1+2\cos\theta}{5+4\cos\theta} d\theta$$

Soln

from $\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx$

when $f(2a-x) = f(x)$ when $f(x)$ is even

ie $I = \int_0^{\pi} \frac{1+2\cos\theta}{5+4\cos\theta} d\theta = \frac{1}{2} \int_0^{2\pi} \frac{1+2\cos\theta}{5+4\cos\theta} d\theta$ as $\cos(2\pi-\theta) = \cos\theta$

$$= \frac{1}{2}$$

= Real part of $\frac{1}{2} \int_0^{2\pi} \frac{1+2e^{i\theta}}{5+4\cos\theta} d\theta$

= Real part of $\frac{1}{2} \int_0^{2\pi} \frac{1+2e^{i\theta}}{5+4\cos\theta} d\theta$

$$z = e^{i\theta} \quad dz = i e^{i\theta} d\theta$$

$$d\theta = \frac{dz}{iz}$$

$$\cos\theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

= R. P. O. $\frac{1}{2} \oint_C \frac{1+2z}{5+\frac{4}{2} \left(z + \frac{1}{z} \right)} \cdot \frac{dz}{iz}$

$$|z|=1$$

$$I = \text{Real part of } \frac{1}{2i} \oint \frac{1+2z}{2z^2+5z+2} dz$$

$$= \text{Real part of } \frac{1}{2i} \oint \frac{1+2z}{(2z+1)(z+2)} dz$$

$$= \text{Real part of } \frac{1}{2i} \oint \frac{1}{z+2} dz \quad \text{--- (2)}$$

poles $z+2=0$

$z=-2$, lies outside the $|z|=1$

ie $f(z) = \frac{1}{z+2}$ is analytic inside $|z|=1$

ie $\oint \frac{1}{z+2} dz = 0$, using Cauchy's Integral theo.

from (2)

$$I = \text{Real part of } \frac{1}{2i} \times 0$$

$$\boxed{I = 0}$$

Q Evaluate $\int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta} d\theta$ by contour integral

Ans $\rightarrow \pi/6$

Q $\int_0^{2\pi} \frac{\cos 3\theta}{5-4\cos\theta} d\theta$

Ans:- $\frac{\pi}{12}$

Soln Hint $I = \int_0^{2\pi} \frac{\cos 3\theta}{5-4\cos\theta} d\theta = \text{Real part of } \int_0^{2\pi} \frac{e^{3i\theta}}{5-4\cos\theta} d\theta$

$$Q \quad \int_0^{2\pi} \frac{d\theta}{5-3\sin\theta} = \frac{-3\pi}{4}$$

$$Q \quad \int_0^{2\pi} \frac{4 d\theta}{5+4\cos\theta} = \frac{8\pi}{3}$$